Practice workbook · Explain ultrafiltration and selective reabsorption in the nephron

High pressure in the glomerulus forcing small molecules into the capsule, what stays in the blood, glucose reabsorption, and why diabetes or damage puts protein or glucose into urine.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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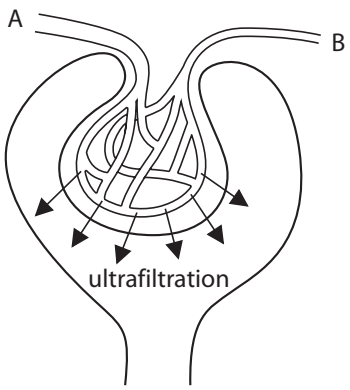
Type ID: HBI-KIDNEY-FILTRATION-REABSORPTION

1. Question 6 · part a.iii · 4 marks

June 2021 Paper 1

Worked-example retry · Attempt part a.iii only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

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1. Question 6 · part a.iii · 4 marks

June 2021 Paper 1

Worked-example retry · Attempt part a.iii only.

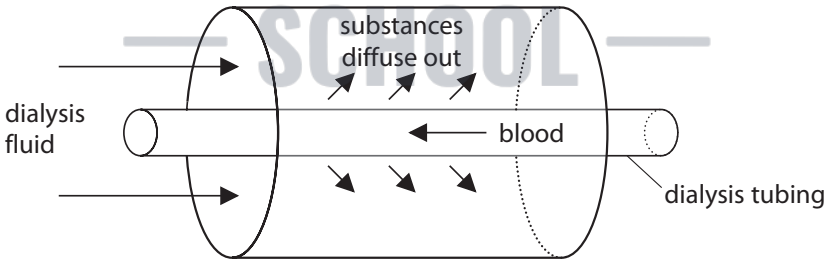
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(iii) Explain how blood vessel A and blood vessel B help the process of ultrafiltration.

(4)

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(b) The diagram shows part of a kidney dialysis machine.



The dialysis fluid contains glucose at the same concentration as in the blood flowing through the dialysis tubing.

Suggest the effect on a person undergoing dialysis if the concentration of glucose was below that of the blood flowing through the dialysis tubing.

(3)

(Total for Question 6 = 12 marks)

1. Question 6 · part a.iii · 4 marks · Working and answer

Working and answer

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2. Question 4 · part a.i · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a.i only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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2. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a.ii · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a.ii only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

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Use the information from the table to answer these questions.

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What is the reason for this? **(1)**

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3. Question 4 · part a.ii · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a.ii only.

(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

percentage = ... %

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3. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

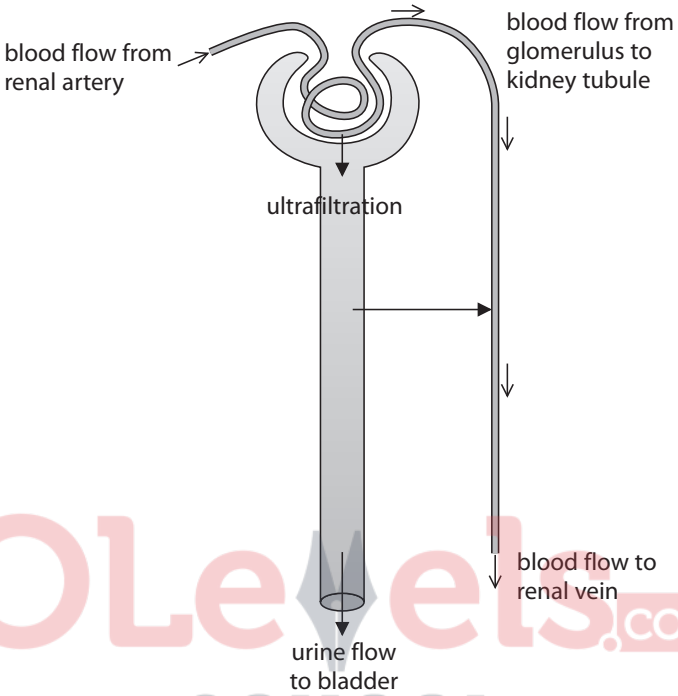
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4. Question 8 · part a.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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4. Question 8 · part a.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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4. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

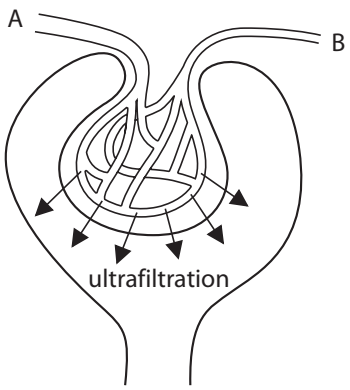
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5. Question 6 · part a.i · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

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(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

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5. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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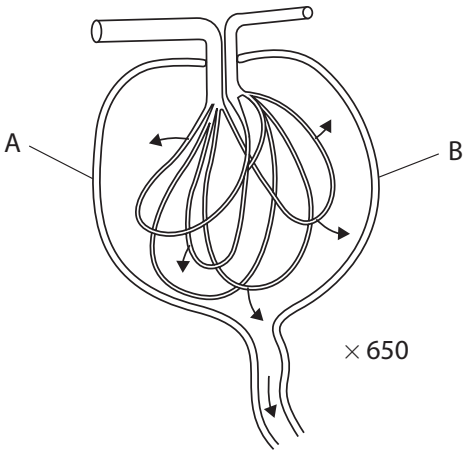
6. Question 1 · part a.iii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.iii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

6. Question 1 · part a.iii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.iii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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6. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

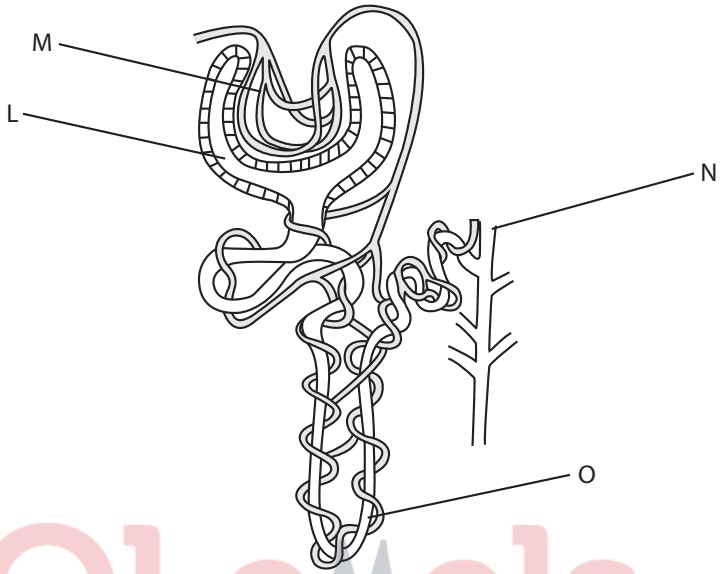
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7. Question 6 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O. **(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L. **(3)**

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7. Question 6 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

7. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

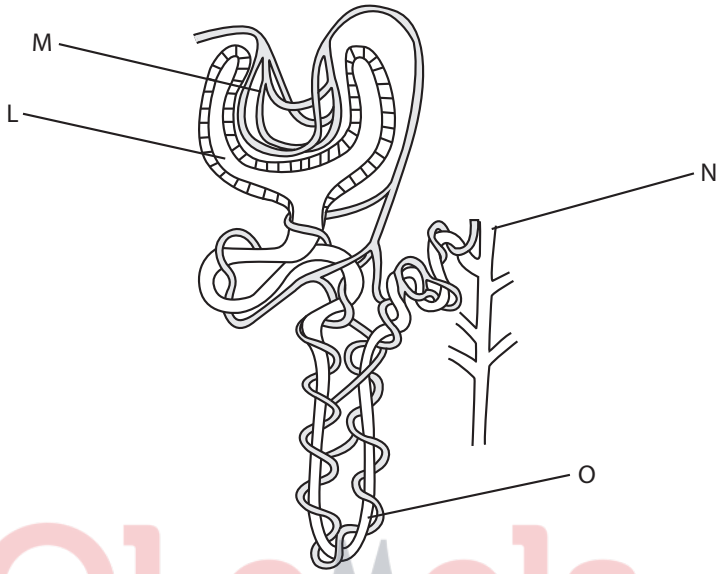
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8. Question 6 · part b.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O. **(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L. **(3)**

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8. Question 6 · part b.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

8. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 5 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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9. Question 5 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

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(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

9. Question 5 · part c · 2 marks · Working and answer

Working and answer

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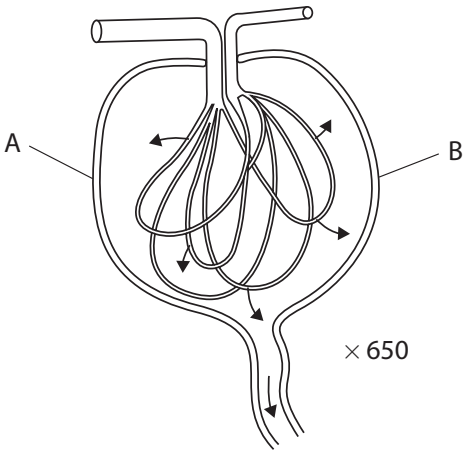
10. Question 1 · part a.iv · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iv only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

10. Question 1 · part a.iv · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iv only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

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Time after drinking in hours	Volume of urine in cm^3	
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4.0	100	50

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10. Question 1 · part a.iv · 1 mark · Working and answer

Working and answer

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11. Question 4 · part a.iii · 3 marks

June 2019 Paper 1R

Further practice · Attempt part a.iii only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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11. Question 4 · part a.iii · 3 marks

June 2019 Paper 1R

Further practice · Attempt part a.iii only.

(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

percentage = ... %

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11. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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12. Question 3 · part c.iii · 3 marks

June 2023 Paper 2

Further practice · Attempt part c.iii only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

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(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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12. Question 3 · part c.iii · 3 marks

June 2023 Paper 2

Further practice · Attempt part c.iii only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

(1)

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

(2)

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(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

(3)

(Total for Question 3 = 13 marks)

12. Question 3 · part c.iii · 3 marks · Working and answer

Working and answer

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13. Question 9 · part c · 4 marks

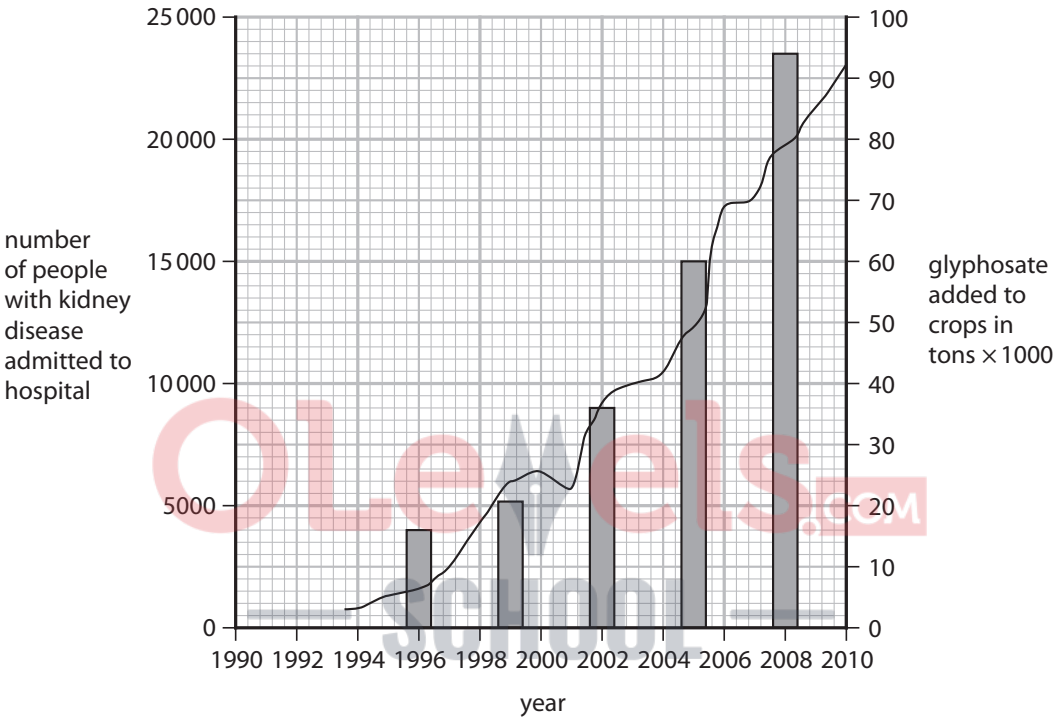
November 2020 Paper 1R

Further practice · Attempt part c only.

9 (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.



(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

(3)

percentage increase = ... %

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13. Question 9 · part c · 4 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

(1)

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(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

(4)

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13. Question 9 · part c · 4 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

(c) There is evidence to show that glyphosate causes inflammation of the glomeruli and the kidney tubules.

Suggest how these symptoms will affect the normal functioning of the kidney and the waste product that it produces.

(4)

(Total for Question 9 = 12 marks)

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13. Question 9 · part c · 4 marks · Working and answer

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